

Emotional Indicators: Uncovering Fake News in Tweets via Emotion Detection

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ABSTRACT

Social platforms are a primary source of news, but also facilitate misinformation. Anger and other negative emotions help misinformation spread. Traditional fact-checking is no longer enough, so automated detection is needed. This must address emotional manipulation and the evolving nature of misinformation. In this paper, we fine-tune three transformer models to improve emotion detection and fake news identification. Additionally, a Gradient Boosting model is trained as a baseline for fake news detection. Our approach focuses on incorporating emotional features into fake news detection, as emotional cues, particularly low arousal levels, appear to be important indicators of misinformation. Our results show that RoBERTa performs well in detecting valence, while BERT achieves strong outcomes in fake news detection, surpassing the Gradient Boosting model in all key metrics.

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1 INTRODUCTION

While misinformation is not a new phenomenon, the 2016 U.S. presidential election marked a pivotal moment in understanding the significant role that fake news plays in shaping political discourse, particularly through social media platforms [4, 10, 22]. Fake news, defined by Allcott and Gentzkow [4] as “news articles that are intentionally and verifiably false and could mislead readers,” extends its disruptive influence well beyond politics. It has been shown to have damaging effects across public health [47], financial markets [14, 30], technology [3], and social cohesion [8, 21]. Moreover, its erosion of trust in reliable news sources and democratic institutions fosters widespread skepticism, threatening societal stability [45]. The growing reach and impact of fake news underscore the urgent need for effective countermeasures.

Social media’s role in the dissemination of fake news is particularly concerning, as these platforms have become a dominant news source for many. Over half of U.S. adults now rely on social media

for news consumption [12]. However, this reliance has contributed to the rapid spread of misinformation, with fake news on social media circulating faster and reaching wider audiences than factual content, especially when emotionally charged [61]. Social media algorithms, designed to maximize user engagement, inadvertently prioritize sensational and emotionally engaging content, often amplifying false information [31, 38]. Emotional engagement is key, as users are more inclined to interact with content that triggers strong emotional reactions, regardless of its accuracy [41].

Emotional content plays a critical role in the virality of fake news. Research indicates that emotions—especially anger—are significant drivers of sharing behavior on social media, with fake news often containing more negative emotions than factual news [13, 26]. Content that evokes strong emotional responses is more likely to be shared, further exacerbating the spread of misinformation. Individuals relying on these emotional cues are not only more susceptible to believing fake news, but also more likely to share it [37]. This emotional manipulation is a deliberate strategy in the construction of fake news, where emotionally evocative headlines and negative sentiments increase both belief in and dissemination of false information [40].

Given the speed and scale at which fake news spreads on social media, traditional fact-checking methods, which rely on human intervention, are insufficient. There is an urgent need for automated systems that can detect fake news at scale and in real time [54]. These systems leverage machine learning algorithms to analyze vast amounts of online content, distinguishing between fake and legitimate news based on features such as linguistic patterns, metadata, and social context [65].

One of the primary difficulties of automated detection systems lies in distinguishing fake news from legitimate content. Fake news often blends elements of truth with misleading narratives, complicating the detection process for algorithms, which must differentiate between satire, propaganda, and falsehoods [15]. Additionally, the rapid evolution of misinformation creates what is known as “concept drift,” where new information renders previously trained models obsolete, necessitating continuous updates that are both costly and labor-intensive [9].

To enhance fake news detection, we propose combining emotional analysis with traditional detection methods. First, we construct an emotionally labeled dataset of tweets based on the Circumplex Theory [48], which will serve as the foundation for training an emotion detection model utilizing the RoBERTa architecture. Concurrently, we develop a traditional fake news detection model. The core contribution of our work is the combination of these two models—emotion detection and fake news detection—to explore

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how dimensionally mapped emotions can improve the classification of fake news. Finally, we will conduct a thorough evaluation of the performance of our emotion detection model, assessing its effectiveness in comparison to traditional methods. This approach aims to bridge a critical gap in the literature by incorporating multidimensional emotional data into fake news detection, a promising avenue suggested by Liu et al. [35]. The Implementation can be found on the following Github Repository¹.

2 RELATED WORK

2.1 Fake News Detection

Fake news detection (FND) methods can be broadly categorized into content-based approaches, feedback-based approaches leveraging user responses, and intervention-based solutions aimed at identifying and mitigating the spread of false information [51, 54]. Content-based methods primarily focus on analyzing the linguistic features, such as syntax, semantics, and style, of the news articles to identify deceptive information [65]. These methods often employ machine learning classifiers, such as Support Vector Machines (SVMs) and Decision Trees, trained on manually labeled datasets of fake and real news [2]. Feedback-based methods, in contrast, leverage user interaction data such as likes, shares, and comments to identify suspicious content. Research has shown, that patterns of user engagement often differ significantly between real and fake news, allowing for social context-based identification [57]. Lastly, intervention-based methods attempt to curtail the spread of fake news by utilizing algorithms to flag suspicious content and by deploying fact-checking tools, helping users critically assess the validity of the information they encounter [42]. These techniques, while foundational, often struggle with the complexity and context-dependence of misinformation, leading to the development of more advanced methods using deep learning models [54, 55, 65].

2.1.1 Transformers in FND. Since the introduction of transformer based architecture by Vaswani et al. [59], Large Language Models (LLMs) like GPT-3.5 and transformers such as BERT (Bidirectional Encoder Representations from Transformers) and have gained prominence in FND tasks due to their ability to capture nuanced semantic and syntactic features in text. These models are pre-trained on vast corpora of data, enabling them to understand language patterns effectively. For example, **BERT-based models** have demonstrated significant performance improvements in fake news detection tasks by capturing contextual relationships within text [18]. Similarly, **RoBERTa**, a refined version of BERT, has shown even better results by optimizing pretraining settings [34].

2.1.2 Fine-Tuning Models in FND. Fine-tuning is essential for adapting LLMs to specific tasks, such as FND. In this process, a pre-trained model is retrained on a smaller, task-specific dataset, allowing it to learn patterns relevant to fake news while retaining its general language understanding [27]. This approach enables models like BERT and RoBERTa to perform well in domain-specific scenarios by combining general and task-specific knowledge [56].

2.1.3 Overview of FND Datasets. The availability of diverse datasets is crucial for advancing FND. Several notable datasets have been

developed, each contributing valuable resources for training and evaluating detection models. However, existing datasets often suffer from limitations related to granularity, modality, size and relevance to current trends [44]. Many of these datasets are also outdated, presenting challenges in maintaining accuracy in a rapidly evolving online information environment [44].

While some datasets encompass data from various domains, a significant proportion is highly domain-specific, limiting their broader applicability. For instance, datasets often focus on particular areas such as celebrity gossip [57], health [16], technology [5], and politics [53]. This domain-specific nature can hinder the development of models capable of generalizing across different contexts, thus requiring broader and more inclusive datasets.

Among the most frequently used datasets for FND tasks are LIAR [62] and the FNC-1 dataset [46]. These datasets have become standard benchmarks in the field. However, TruthSeeker, introduced in 2023, offers a significant update, with over 180,000 labeled tweets specifically targeting fake news on social media, making it one of the largest and most relevant datasets for modern FND research [17].

2.2 Text Classification using GPT

Text classification, sentiment analysis, and emotion detection are interrelated tasks within natural language processing. Text classification encompasses the broader process of assigning predefined labels to textual data [49], with sentiment analysis and emotion detection being specialized subfields. Sentiment analysis (also known as 'Opinion Mining') focuses on determining the overall polarity of a text—whether it expresses positive, negative, or neutral sentiment [39, 63]. In contrast, emotion detection provides a more granular understanding by identifying specific emotions such as joy, anger, or sadness, often relying on multidimensional emotion models [36, 50].

A modern approach to prompt engineering tailors input structures to ensure the model retrieves the most accurate results. This technique, which has evolved alongside large pre-trained models like GPT, emphasizes adjusting prompt phrasing and examples to improve output quality and relevance [64]. Whether constructed manually or via automated processes, prompt engineering plays a pivotal role in adapting foundation models for a wide range of applications [23, 32]. There are a few notable approaches[11]:

- **Zero-Shot Classification:** In zero-shot classification, the model performs a task based solely on natural language instructions without being trained on any specific examples of the task. This method can generalize to unseen categories.
- **Few-Shot Classification:** Few-shot learning involves providing a model with a small number of task-specific examples during inference. This reduces the need for large datasets, but still relies on minimal task-specific data.

2.3 Emotion Theories

Emotion theories are generally classified into two main categories: categorical models, which directly define discrete emotions, and dimensional models, which represent emotions within a multi-dimensional space. A widely used categorical model in emotion detection, often referenced in studies such as [6, 7, 29], is the Ekman

¹https://github.com/NoahMeissner/Fake_News_Emotions_Detection

model, which identifies six fundamental emotions: anger, disgust, fear, joy, sadness, and surprise [19]. Other categorical models vary in the number of core emotions, typically ranging between 6 and 10, as discussed by Izard [28], Shaver et al. [52], and Tomkins and McCarter [58]. Among the notable dimensional models are Plutchik's Wheel of Emotions, which organizes eight primary emotions in a circular structure [43], the Lövheim Cube of Emotions, which represents emotions across three biochemical dimensions (noradrenaline, dopamine, and serotonin), and the Circumplex Model proposed by Russell [48], where emotions are placed in a two-dimensional circular space with valence (ranging from negative to positive) and arousal (ranging from low to high) as the axes.

2.4 Combining FND and Emotion Detection

There is evidence in the literature suggesting that incorporating emotional features could improve fake news detection. For instance, Luvembe et al. [36] demonstrated that integrating dual emotion features, specifically publisher and social emotions, significantly enhances detection accuracy.

Additionally, research has established a clear link between emotions and fake news. Fake news titles tend to convey stronger emotions and exhibit less neutrality compared to real news titles [24, 25, 40]. Furthermore, fabricated news, both in titles and article bodies, is marked by significantly higher levels of negative emotions, particularly anger and disgust, and substantially lower levels of joy compared to real news content [40].

However, these findings are constrained by the study's reliance on a categorical model, which only included the emotions of joy, fear, disgust, and anger. In contrast, we argue that employing Russell's Circumplex Theory allows for a more nuanced and comprehensive analysis of emotional expression in tweets. This model captures a broader spectrum of emotions by mapping them along the dimensions of valence and arousal, offering a more precise and in-depth understanding of emotional patterns in the context of fake news detection.

3 METHODOLOGY

This section provides a comprehensive account of the selection of materials and the methodology employed to implement the chosen approach. The project is divided into three steps. The following section elucidates the process of annotating 50,000 tweets with regard to their emotional mood using GPT-3.5 and a few-shot approach. Subsequently, the fine-tune classification models for valence, arousal and fake news are analysed and explained in more detail. The final part is dedicated to the training of a Gradient Boosting Model for the classification of fake news on the basis of emotions.

3.1 Material

In order to gain insight into the dissemination of misinformation on social media, the Truth Seeker 2023 data set was employed [17]. This comprises 180,000 tweets from the platform X (formerly Twitter) that have been classified as pertaining to fake news. The data was initially sourced from the Politifact dataset [60] and subsequently extracted through crawling and subjected to further processing. The dataset comprises 1,400 statements, which have been divided

equally into two categories: those that are truthful and those that are fake². As figure 1 shows, the dataset was divided into different sub-areas, allowing a substantial amount of data to be used for training while ensuring an adequate amount was available for evaluation. This allows us to include statements in the evaluation that do not appear in the training data.

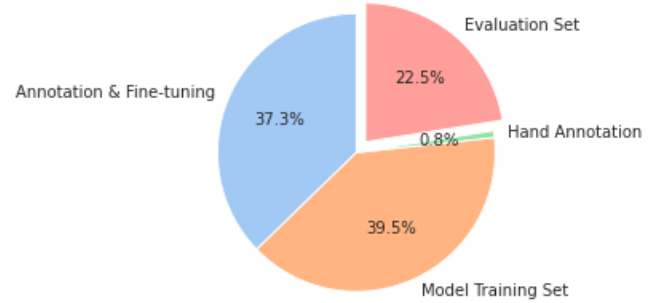


Figure 1: Dataset Distribution

3.1.1 Pretrained Language Models. BERT: In developing our fake/truth classification model, we used BERT, a pretrained encoding model[18]. BERT uses masked language modelling and a self-attention mechanism that allows us to fine-tune it by adding an additional output layer specifically tailored to our task[18].

RoBERTa: For the classification of emotions we used RoBERTa as base model, which builds on BERT and is characterised by significant improvements in terms of contextual understanding[33]. In particular for the Evaluation process, the model roberta-base-go_emotions was used, which was trained for the classification of 28 different emotions³.

GPT-3.5 TURBO: The annotations were created using the few-shot method in conjunction with GPT-3.5 Turbo⁴. Therefore we tested as well other Models of the GPT Family.

RoBERTa-base-go_emotions The RoBERTa-base-go_emotions model⁵ was used for the evaluation process. The model is based on RoBERTa and was adapted to encompass 28 distinct emotion categories through the addition of an additional classification layer. The GoEmotions dataset, which is based on Reddit data, was used as training data⁶. In order to facilitate comparison of the results produced by our Emotion Detection models, the emotion categories were mapped to the valence-arousal model as shown by Liu et al. [35]. This approach enables the model to serve as a reference point for evaluating the performance of our own model in emotion recognition.

3.2 Experimental Setup

The project uses the Schlaubox environment⁷ and Python. For the fivefold cross validation of our fine-tuned models, we used Google

²<https://www.unb.ca/cic/datasets/truthseeker-2023.html>

³https://huggingface.co/SamLowe/robertabasego_emotions

⁴<https://platform.openai.com/docs/models/gpt-3-5-turbo>

⁵https://huggingface.co/SamLowe/roberta-base-go_emotions

⁶https://huggingface.co/datasets/google-research-datasets/go_emotions

⁷<https://schlaubox.de/>

Example 1: Tweet: "@newsmax @bennyjohnson Biden plans to tax your 401K. Believe it. The democrats are spending so much money they need new revenue sources. Wake up America. This is YOUR MONEY!!!!!" Example 2: Tweet: "Congrats to Ivy Taylor, first African American mayor of San Antonio."	
Valence: Negativ Valence: Neutral Valence: Positiv	Arousal High Arousal Medium Arousal Low

Figure 2: Annotation Examples

Collab PRO⁸ as environment⁹. The complete project, including a comprehensive description. Pandas¹⁰ and Numpy¹¹ were utilized for data processing, while SciKit¹² and xgboost¹³ was employed for machine learning operations. The transformer library¹⁴ and PyTorch¹⁵ were utilized for NLP tasks. For Evaluating purposes we used pingouin¹⁶ and scipy¹⁷.

3.2.1 Data Preperation. In the first step, the entire dataset was split into a training set and an evaluation set in an 80:20 ratio, as shown in Figure 1. This split was based on the tweet topics. The training data set was then used for further processing. Next, the training data was divided into two parts. The first part was used to create an annotation pipeline and to fine-tune a BERT classification model for detecting fake news, as well to fine-tune a RoBERTa classification model for Valence and Arousal Detection. The second part was used to develop and optimize a gradient-boosting model for fake news detection.

3.2.2 Emotion Labelling. Utilizing the Circumplex Model [48], we classified each sequence according to the dimensions of Arousal (High, Medium, Low) and Valence (Positive, Neutral, Negative). These classification categories were adapted from the approach outlined by Liu et al. [35]. The annotation of the 50,000 tweets was carried out in three stages. At the beginning of the project, a codebook was developed to guide the human annotators in making correct classification decisions¹⁸. In the second step, 1,040 tweets were categorised by two independent annotators, as shown in figure 2. Following this, a discussion was held to address any discrepancies in the annotations, and a consensual classification was determined. In the next stage of the process, the same scales were used by the GPT-3.5 Turbo and GPT-4o-mini models to annotate the tweets. The implementation was guided by the methodology described in by Achmann-Denkler [1]. The kappa value [20] was used as the evaluation metric, with GPT-3.5 Turbo being identified as the most effective tool, as show in table1. In the third and final step, the text data was then annotated using GPT-3.5 Turbo¹⁹.

⁸<https://colab.research.google.com/>

⁹Training cost: \$12.9

¹⁰<https://pandas.pydata.org/>

¹¹<https://numpy.org/>

¹²<https://scikit-learn.org/>

¹³<https://xgboost.readthedocs.io/en/stable/>

¹⁴<https://huggingface.co/docs/transformers/index>

¹⁵<https://pytorch.org/>

¹⁶<https://pingouin-stats.org/build/html/index.html>

¹⁷<https://scipy.org/>

¹⁸<https://shorturl.at/QhbKd>

¹⁹The cost for 50,000 tweets were \$28.12

Model / Annotator	Arousal	Valence	Together
Annotators	0.6	0.43	0.38
GPT 3.5 Turbo	0.83	0.74	0.72
GPT 4o - mini	0.46	0.66	0.43
(RoBERTa Emotions)	0.006	0.09	0.03

Table 1: Kappa Values

3.2.3 BERT Fine-Tuning for Classification. As part of the data segmentation described in the section 3.1, the tweets were used for the training process without additional pre-processing, as the data set already provides them in a suitable format. To further optimise the detection model based on BERT, an additional classification layer was integrated. The model was then trained on n = 45,000 training examples, comprising 3 epochs, with a learning rate of 4e-5, a weight decay of 0.01 and 500 warm-up steps.

3.2.4 RoBERTa Fine-Tuning for Emotion Classification. The RoBERTa-Base model was used to apply the arousal and valence recognition models. The data set was generated from the section on emotion annotation, Section 3.2.2, which included n = 50,000 data points. Prior to modeling, the data was pre-processed to ensure that only the correct categories were used. Standardised training parameters were defined for both models. These comprised training over 3 epochs, with a learning rate of 1e-5, a weight decay of 0.005, 200 warm-up steps and a label smoothing factor of 0.1.

3.2.5 Definition of the Gradient-Boosting Models. Gradient-boosting models were developed to detect instances of fake news, using emotional annotations as inputs. The maximum tree depth (max_depth) was set at 3, the learning rate (learning_rate) at 0.1, and the number of trees (n_estimators) at 100, for all models. The models were constructed with differing selections of training variables and training data. In the preprocessing, the data was first pre-classified using the RoBERTa-Valence and RoBERTa-Arousal models to predict emotion labels. These predicted labels were then used as features for training the models.

- **Model 1:** This model was trained on the entire training set.
- **Model 2:** This model was also trained using the emotion data, but without the 25 % shortest text sequences..

The variability introduced to the training set permits an investigation into the influence of data selection criteria on the efficacy of the model in detecting fake news.

4 EVALUATION

4.0.1 Evaluation of the Annotationprocess. The annotation pipeline was evaluated using a corpus of 1,040 test tweets. The manually annotated ground truth served as a reference point for assessment. To validate the models, the kappa coefficient was employed to assess the degree of agreement between the model results and the ground truth 1. To compare those values, a z-Test was conducted to calculate the differences.

4.0.2 Evaluation of the fine-tuned Models. The performance of the models was evaluated both during training (after each epoch) and after the completion of training. As part of the evaluation, cross-validation was also carried out in order to determine metrics such as

the F1 score and accuracy on the basis of the validation split. In view of the use of unbalanced data, the F1 score proves to be decisive. A classification report was created as part of the comprehensive analysis of the model results.

Furthermore, a five-fold cross-validation of the fine-tuned models was performed to evaluate the performance of the model. The outcome were compared using paired t-tests to identify significant differences between them. Furthermore, the fine-tuned fake news detection model was tested with the evaluation data and compared with these results.

To compare the fine-tuned RoBERTa models with the external RoBERTa emotion model, these models were used to classify the manually labeled dataset split. Kappa coefficients were calculated to assess the agreement between the models and the manually labeled classifications, and a z-test was performed to determine if there was a significant difference in their performance.

4.0.3 Gradient Boosting Models. In this model category, the results were cross-validated after training, and a classification report was generated to support analysis. Metrics such as precision, recall, F1 score, and accuracy were calculated using the validation set. Given the imbalanced dataset, the F1 score was particularly important. As outlined in Section 4.0.2, a 5-fold cross-validation was performed to compare the Gradient Boosting Models with each other and the fine-tuned Fake News Detection Model, using a paired t-test for evaluation.

5 RESULTS

5.0.1 Explorative Analyse. Figure 3 illustrates the distribution of emotions in the categories, divided into fake and true news. The diagram shows a clear uneven distribution, particularly in the area of negative valence. The combination of negative valence and high arousal is of particular relevance here. Furthermore, the graph reveals significant differences in the area of low arousal between fake and true news. These observations lead to the conclusion that the data set shows a certain imbalance, with the focus being on negative emotions with high arousal.

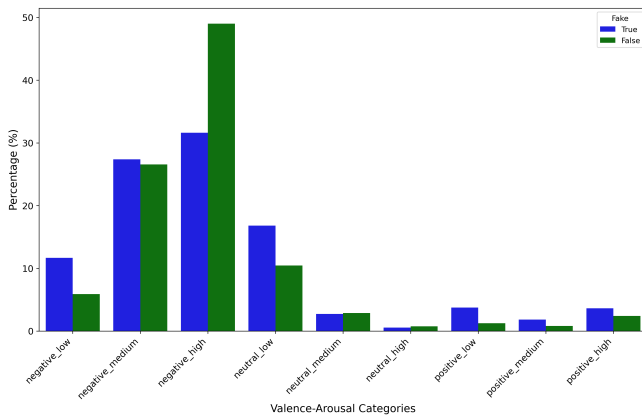


Figure 3: Combined Valence-Arousal Distribution

5.0.2 Results of the Annotation Process. As part of the evaluation process, kappa coefficients were determined, which demonstrate

Models	Accuracy	Precision	Recall	F1
RoBERTa-valence	0.86	0.73	0.74	0.74
RoBERTa-arousal	0.71	0.70	0.70	0.7
(RoBERTa-Emotions)	0.47	0.58	0.40	45

Table 2: Emotions Models

substantial agreement between the ground truth and GPT-3.5 Turbo for the dimensions of Valence ($\kappa = 0.74$), Arousal ($\kappa = 0.83$), and combined ($\kappa = 0.72$) as seen in Table 1. This indicates that GPT-3.5 Turbo is capable of accurately reproducing the annotated data, as well as that there is a high degree of agreement between the model and the ground truth.

In contrast, the comparison model, GPT-4o mini, exhibits a notable discrepancy in performance. The agreement values are moderate, with a low correlation observed for the dimensions 'Valence' ($\kappa = 0.66$), 'Arousal' ($\kappa = 0.46$), and 'Combined' ($\kappa = 0.43$), Table 1. The Z-test shows a significant difference in performance between the two models ($Z = 11.57$, $p < 0.001$), which emphasises the considerable influence that the choice of model can have on the agreement with the ground truth.

5.0.3 Results of the Emotion Detection Models. Table 2 summarises the performance metrics of the various detection models. Due to the imbalance of the training data, it is necessary to use the F1 score in order to make valid statements. With an accuracy of 0.86, a precision of 0.73, a recall of 0.74, and an F1 score of 0.74, the RoBERTa-valence model shows moderate results. The RoBERTa-arousal model shows similar results in terms of accuracy (0.71), precision (0.70), recall (0.70), and F1 score (0.70). To analyse possible differences between these models, a paired t-test was conducted ($F = 8.77$, $p < 0.001$). The test showed that there is a significant difference in the performance.

However, significant differences in the test results were found between the RoBERTa-valence model ($Z = 16.1$, $p < 0.001$) and the RoBERTa-arousal model ($Z = 18.6$, $p < 0.001$) compared to the external RoBERTa-emotions model. This leads to the conclusion that the self-trained models perform significantly better than the external RoBERTa-emotions model. Furthermore, the test evaluation dataset of all three models was classified, and the kappa value between the models was calculated. The results ($Z = 18.5$, $p < 0.001$) also show significant differences between the Valence, Arousal, and RoBERTa-emotion models.

5.0.4 Results for the Fake News Detection Models. As shown in Table 3, the BERT-Finetuned model achieves excellent performance in the detection of fake news, with high accuracy (0.99), precision (0.99), recall (0.99), and F1-score (0.99). These results illustrate the high performance of the model in the first evaluation step.

In a second step, the model was evaluated on an independent dataset, which showed an observable decrease in the test statistics, with precision (0.77), recall (0.82), accuracy (0.79), and F1-score (0.71). The results on the evaluation dataset are moderate compared to the training validation dataset, as reflected in the accuracy, precision, recall, and F1-score. These deviations suggest a limited generalisability of the model, especially when applied to new subject areas.

Furthermore, two gradient-boosting models with divergent training data were defined. The first model was trained on the whole training set and showed moderate results with accuracy (0.61), precision (0.62), recall (0.64), and F1 (0.63). The second model was trained on the same training data, excluding 25% of the items with the shortest text sequences. Like Model 1, we received moderate results with accuracy (0.61), precision (0.54), recall (0.55), and F1 (0.57). The comparison with a paired t-test shows ($F = 15.32$, $p < 0.001$) that Model 1 outperforms Model 2.

As part of a further investigation, the evaluation data was classified with both models. Table 3 shows no observable difference between the validation metrics and the evaluation metrics.

The comparison between the BERT-Classification Model and the first Gradient Boosting Model revealed a significant difference in performance between the fine-tuned BERT model and the emotion gradient boosting model, as shown by the paired t-test ($F = 78.13$, $p < 0.001$).

Modela	Accuracy		Precision		Recall		F1
BERT	0.99	0.79	0.99	0.77	0.99	0.82	0.99 0.79
GB Model 1	0.61	0.60	0.62	0.60	0.64	0.61	0.63 0.61
GB Model 2	0.61	0.60	0.54	0.60	0.55	0.62	0.57 0.61

Table 3: Detection Models [Training Data | Evaluation Data]

6 DISCUSSION

The emotion detection results indicate moderate to good performance in recognising and classifying them. In combination with the comparison of external models, the results illustrate a robust performance that provides a solid basis for further work. At the same time, however, there is also potential for improvement, particularly in the area of arousal detection. The results indicate that the models have difficulties reliably classifying tweets in terms of their emotional arousal.

However, the classification of fake news based on the recognised emotions provides less convincing results. The exploratory analysis, in particular the distribution of emotions, indicates an imbalance in the dataset. For both fake news and true news, a high number of emotions were observed in the ‘negative high’ category. Furthermore, the analysis of the training parameters reveals that the arousal low category in particular could be relevant for the model.

With regard to the BERT detection model, a significant impairment in the model’s performance can be observed due to the content limitations of the training data set. This indicates that the integration of additional parameters could significantly improve the robustness and performance of the model.

The comparison between the BERT classification model and the Boosting Gradient Models, shows as well that the performance of BERT outperforms the Emotion Detection approach.

6.0.1 Limitations. Overall, the annotation pipeline shows a substantial agreement and performs well in the classification of political tweets. Nevertheless, it should be noted that the use of alternative models, such as GPT-4, could potentially improve performance, especially in terms of context understanding. Such an optimisation of the annotation results as well as an increase in analysis quality can be expected.

An additional critical consideration is the unique structure of tweets, particularly their brevity. The short length of many tweets presents a challenge for both machine learning models and human annotators in terms of accurate classification. This problem can affect the reliability and consistency of the results, which is why it should be taken into account when interpreting the data.

Furthermore, the data set is characterised by low variability due to the selective choice of exclusively political tweets from the USA. This limitation means that the results can only be applied to the specific topic area of Twitter. This represents a further limitation, as the generalizability of the results to other contexts or subject areas is therefore considerably restricted.

Moreover, a significant, uneven distribution of emotions can be observed within the data set. The distribution shown is the result of data collection by the providers and cannot necessarily be transferred to the actual emotional diversity in political communication on Twitter. The question whether the emotions are actually distributed in the same way or whether the inequality is due to the crawling process remains open.

7 CONCLUSION AND FUTURE WORK

In this paper, we examine the process of fine-tuning three transformer models. The RoBERTa model was employed for the detection of valence and arousal, while BERT was utilised for the identification of fake news. The findings indicate that while the models exhibited robust performance, particularly in valence detection (precision, recall, accuracy, F1 score), there is scope for further enhancement. The RoBERTa model demonstrated superior performance in valence detection, attaining well metrics across all evaluation metrics (precision (0.73), recall (0.74), accuracy (0.86), F1 (0.74)). BERT also exhibited encouraging outcomes in fake news detection, with competitive metrics in precision (0.99), recall (0.99), accuracy(0.99), and F1 score (0.99).

Furthermore, two Gradient-boosting models were trained for the purpose of detecting fake news. However, it delivered moderate performance across the same metrics as the first model shows (precision(0.62), recall(0.64), accuracy(0.61), F1(0.63)). Additionally, this model was outperformed by the BERT classifier in all aspects.

However, a major challenge remains in reliably classifying tweets based on their level of arousal. The results suggest that low arousal levels may be an indicator for detecting fake news, highlighting the potential of this parameter in distinguishing between real and fake news and underscoring its relevance for misinformation detection.

Our results only partially confirm previous research, as discussed in section 2.4, suggesting the need for further optimisation. In addition, the current results appear to have limited generalisability to other platforms. Future research should explore alternative methods, such as using other transformer models for data annotation, or considering the entire conversation as a contextual entity rather than individual tweets. This could help to gain a deeper understanding of context and improve both the annotation process and overall model performance.

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